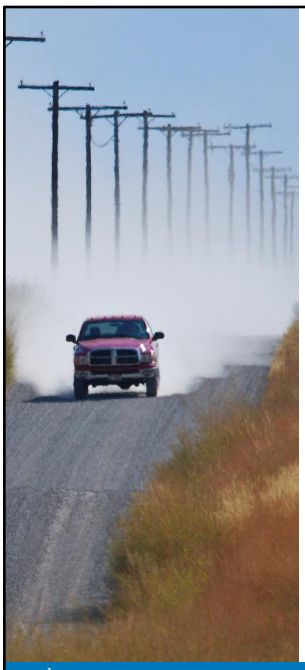


Learning Outcomes

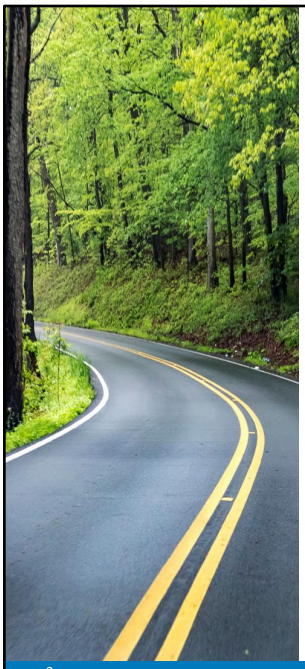
- List the differences between systemic screening and other network screening methods
- Describe the three elements that make up the Systemic Safety Project Selection Tool
- Describe and apply the 4-step systemic planning process
- Identify resources that are helpful in conducting systemic safety analysis

Suggested Comments/Key Message: By the end of this module, you should be able to:

- List the differences between systemic screening and other network screening methods
- Describe the three elements that make up the Systemic Safety Project Selection Tool
- Describe and apply the 4-step systemic planning process
- Identify resources that are helpful in conducting systemic safety analysis

Facilitation Guidance/Interactivity: Review learning outcomes. Note that at the end of the training you will revisit to make sure they have been met. Also consider asking the participants the questions listed below if time allows.

Potential Answers/Sample Q&A:
Ask: Please take a moment and think about what your specific learning outcomes are? Why are you participating in this training? What do you want to get from it? How might some of these ideas be valuable to you in your work?
Answer: All answers to this question are correct. Please jot down your answers in your workbook.



Module Goal

Upon successfully completing this module, participants will be able to describe the benefits of the data-driven, systemic safety project selection tool and how to apply this tool to their transportation networks.

Suggested Comments/Key Message: By the end of this course our goal is that each participant will be able to describe the benefits of the data-driven, systemic safety project selection tool and how to apply this tool to their transportation networks

Suggested Comments/Key Message: Before we discuss the systemic safety project selection tool, we will first introduce systemic safety and discuss how it compares to other network screening methods. This will help us meet our first learning outcome "List the differences between systemic screening and other network screening methods".

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Systemic Approach: Definition

The term "systemic safety improvement" means an improvement that is widely implemented based on high-risk roadway features that are correlated with particular crash types, rather than crash frequency.

-- 23 USC 148 (a)(12) Systemic safety improvement

Source: USDOT

Suggested Comments/Key Message: Before we jump in to describing how to complete a systemic safety analysis, lets first discuss the definition of systemic safety. The following definition is from the US Code. With traditional hot spot approach, we identify locations with a large number of crashes, a high crash rate, or a large number of crashes relative to other similar sites. Rather than looking at locations with crashes, in systemic analysis, we are looking to identify locations with risk factors that are associated with crash types.

Reference Materials: 23 USC 148 (a)(12) Systemic safety improvement

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Systemic Approach

A systemic approach involves the installation of low- to moderate-cost countermeasures at locations with the highest risk of severe crashes.

Source: Walt Catlett, RRSC

Suggested Comments/Key Message: Another way to look at it is that the Systemic Approach involves the installation of low- to moderate-cost countermeasures at locations with the highest risk of severe crashes.

Systemic screening Identifies a “problem” based on system wide data, such as rural lane departure crashes, urban pedestrian crashes, or rural unsignalized intersection crashes. These crashes are often spread across the network with few or no locations experiencing a “cluster” of crashes during a given period of 3-5 years, but which still present a safety risk to the travelling public. The sites where these crashes occurred are then studied looing for characteristics (i.e., geometry, volume, or location) that are frequently present in these severe crashes. These characteristics are referred to as risk factors.

Systemic focuses on promptly deploying one or more low-cost countermeasure to address the underlying circumstance contributing to crashes on most roads sharing a set of risk factors. By addressing crash types experiencing low densities (crashes per intersection or mile) but high aggregate numbers, program funds can be dedicated toward low-cost solutions deployed across the system, affecting many locations. And the process identifies and prioritizes locations across the roadway network for implementation.

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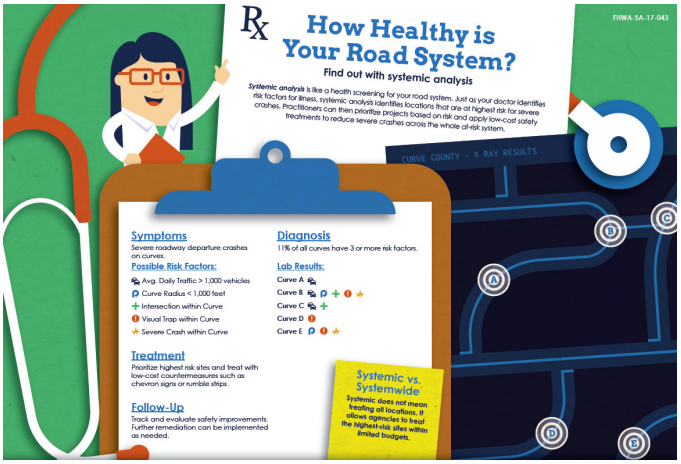
How Healthy is Your Road System?



Source: FHWA

Suggested Comments/Key Message: The systemic safety approach is like going to the doctor's office. Doctors are not going around looking for patients with heart attacks, instead they are screening for factors that are associated with an increased risk of heart disease or other diseases. They then try to treat those patients proactively, and do not just sit around waiting for the patient to have a heart attack before treating. This is what the systemic approach is all about. If we only follow a hot spot approach, we will always be chasing these sites. With the systemic approach we are trying to get out and try to prevent crashes before they occur, just like doctors try to present heart attacks.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/ddsa_systemic_analysis.pdf

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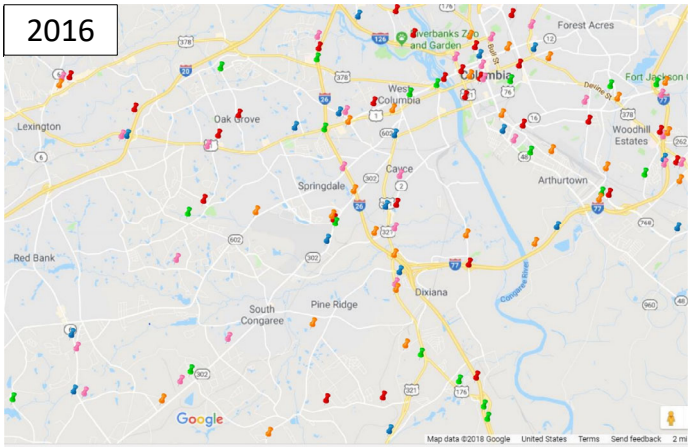
Where would you invest money to improve safety?



Source: Eastern Washington via FHWA

Suggested Comments/Key Message: So why is the systemic approach needed? In the past, the method used to identify locations for improvements was primarily to look at where crashes have occurred. We are going to look through five years of crash data to see if you can identify a location where you might want to invest in an improvement. **(Click 4 times).**

Before us is data from 2012 to 2016. Were you able to identify any sites on the map that have a large number of crashes that you would target the site for improvement? Take a moment and share your thoughts in the chat pod…. *(Review answers placed in chat pod)*

As we can see it is hard to determine sites with a large number of crashes, as overall, the crashes appear to be pretty widespread. Severe crashes often occur at seemingly random locations and tend not to cluster over time, especially in rural, low to moderate traffic volume contexts. If we focus only on high crash locations, we will never make it to zero deaths on our roadways. However, a different approach to improving safety, would be to make improvements based on risk factors, this is called the systemic approach. Severe crashes are the ones that can lend themselves well to a systemic approach.

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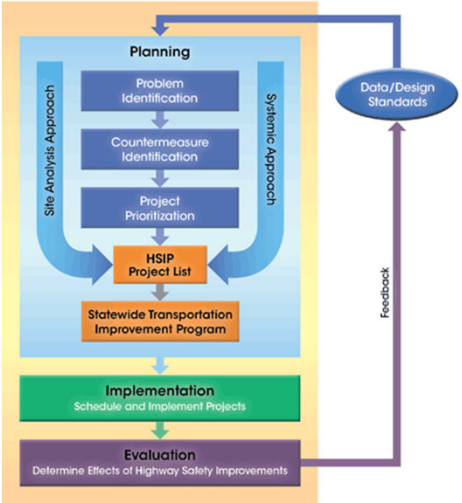
Systemic Approach

- Complementary approach to traditional site-specific analysis
- Primary approach for rural and local roads
  - Can still be applicable to urban roads
- Crashes alone do not establish prioritization



Source: FHWA

Suggested Comments/Key Message: A few points we want to make sure we highlight related to the systemic approach are shown. First, the systemic approach is not meant to completely replace the more traditional site specific (hot spot) approach. These should be used together to address both those high crash locations as well as those sites with risk factors associated with overrepresented crash types. As we can see in the figure on the right these approaches can be used together to determine a list of projects for your Highway Safety Improvement Program (HSIP) funds. Both types of approaches are really needed to help us reach zero deaths on our roadways.

The systemic approach is likely the primary approach for rural and local roads due to the more widespread nature of the crashes seen in those locations. These sites may want to start with the systemic approach and then supplement with a hot spot approach. However, we want to note that the systemic approach is still applicable to urban roads as there are widespread crash types that occur on those roads as well.

One of the main differences between the systemic approach and the site specific approach, crashes alone do not establish prioritization, instead the risk of crashes occurring does.

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Benefits of Systemic Safety Planning

- Proactive program to address likelihood of severe crashes
  - Addresses 'random' nature of severe crashes
- Greater knowledge of severe crashes
  - Contributing factors and location characteristics
  - Improve planning, design, and maintenance practices
  - Risk management for tort liability
- Magnitude of crash reductions
  - Case by case

Suggested Comments/Key Message: There are benefits related to conducting systemic safety planning. The greatest benefit…IF projects are implemented…is the expectation that severe crashes will be reduced.

Additionally, knowledge gained from this analysis approach can help an agency avoid creating the same circumstances observed in severe crashes during their planning, design, and maintenance stages. It can also help manage tort liability risk.

The magnitude of crash reductions expected will vary and aren't widely calculated yet due to this being a relatively new concept, limited statistical documentation is available showing the exact reductions realized from using a systemic approach. However, it is important to recognize several of the states that have implemented systemic improvements have reported positive results in terms of crash reductions since embarking on their systemic safety efforts.

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Definitions

- Traditional Site Specific (Hot Spot)
  - Deploying countermeasures at the ***highest crash*** locations
- Systematic
  - Deploying countermeasures at ***ALL*** locations
- Systemic
  - The installation of low- to moderate-cost countermeasures at locations with the highest risk of ***severe crashes***

Source: Walt Catlett, RRSC

Suggested Comments/Key Message: We have talked quite a bit about the traditional site specific as well as systemic approach today, but we haven't mentioned systematic safety, which often gets confused with systemic. Here are a few definitions related to each of these and how they differ:

- Traditional Site Specific (Hot Spot) is deploying countermeasures at the ***highest crash*** locations,
- Systematic is deploying countermeasures at ***ALL*** locations, and
- Systemic is the installation of low- to moderate-cost countermeasures at locations with the highest risk of ***severe crashes .***

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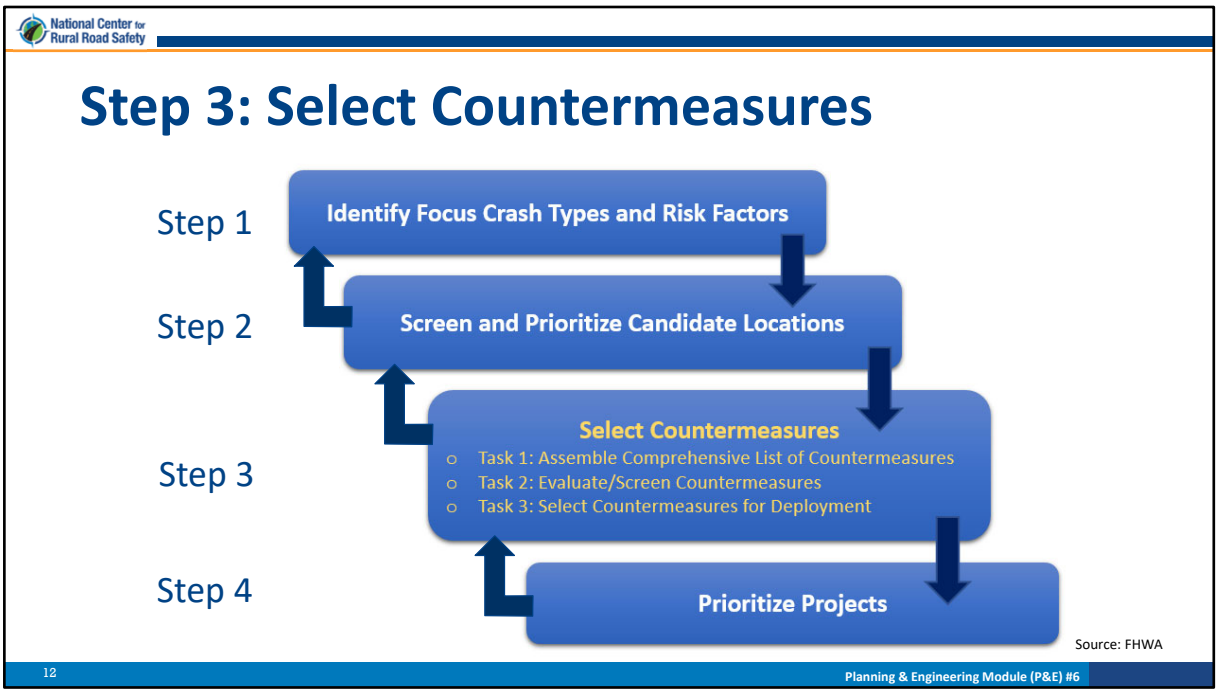
3 Approaches to Safety

https://www.youtube.com/watch?v=1Gtz0qjPx0M

Source: FHWA

Suggested Comments/Key Message: We talked about the definitions on the previous slide, now let's take a look at this FHWA video called "3 Approaches to Safety" further detailing the differences in the three approaches. *I love this video* [*play video*].

Facilitation Guidance/Interactivity: The FHWA video called "3 Approaches to Safety" (https://www.youtube.com/watch?v=1Gtz0qjPx0M) is 4 minutes 31 seconds long. Play the video for the audience.



Suggested Comments/Key Message: Now that you have your prioritized list of candidate locations from step 2, one needs to move onto step 3 where you select countermeasures. The objective of this step is to assemble a small number of low-cost, highly effective countermeasures to be considered for project development at candidate locations identified in the previous step.

This step is again broken down into three tasks and begins with incorporating research into another macro level decision related to the systemwide data analysis results from the first step and moves to a micro level selection based on specific jurisdictional experience, policies, and practices regarding potential countermeasures.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

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Step 3 – Task 1: Assemble Comprehensive List of Countermeasures

- Determine a list of countermeasures that will address the focus crash type and facility
- References to consult:
  - FHWA's Proven Safety Countermeasures
  - CMF Clearinghouse
  - Highway Safety Manual
  - NHTSA's Countermeasures That Work
  - Strategic Highway Safety Plans
  - Intersection Safety Plans
  - Roadway Departure Safety Plans
  - Human Factors Guidelines
  - NCHRP Report 500 series

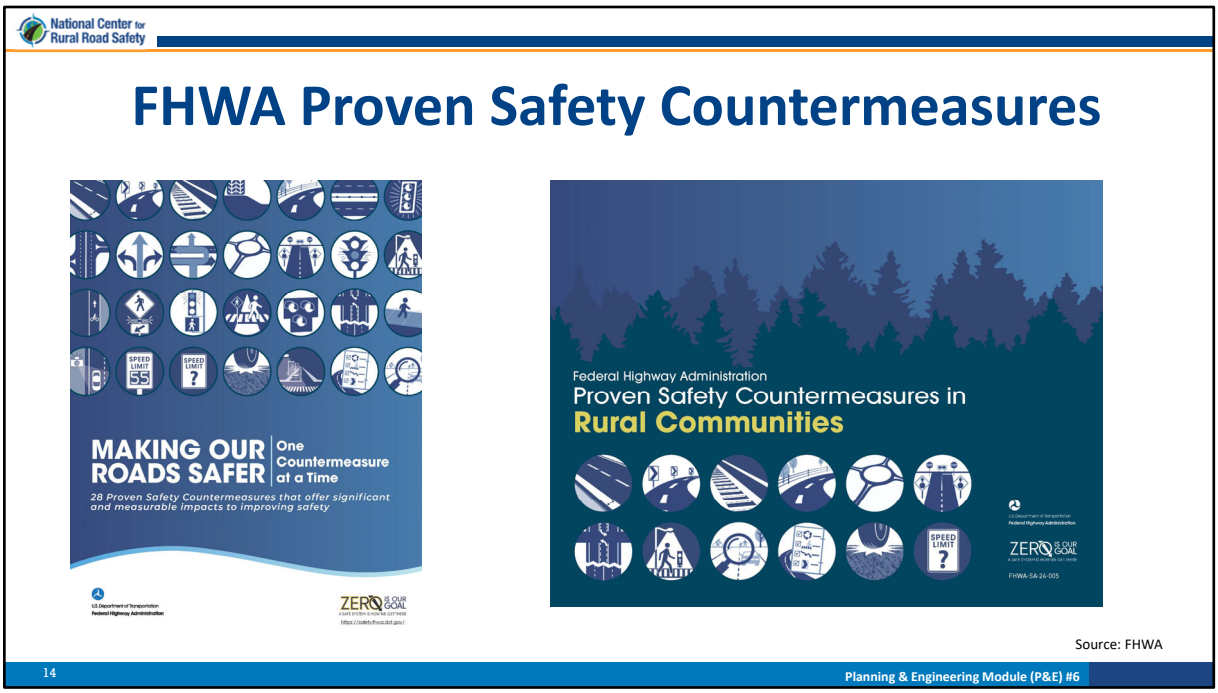
Suggested Comments/Key Message: The first task of step 3 is to assemble a comprehensive list of countermeasures to consider. When developing this list, we want to make sure the countermeasures we are including will address the focus crash type and facility. **(CLICK)** There are a variety of references one can consult when assembling a list of countermeasures.

Facilitation Guidance/Interactivity: n/a

Potential Answers/ Sample Q&A:
***Ask*:** What are some references you would consult? (type in the chat pod)
***Answers*:** See above on slide.

Reference Materials:
FHWA Proven Safety Countermeasures: https://highways.dot.gov/safety/proven-safety-countermeasures
CMF Clearinghouse: http://www.cmfclearinghouse.org/
Highway Safety Manual: http://www.highwaysafetymanual.org/Pages/default.aspx
NHTSA's Countermeasures That Work:
https://www.nhtsa.gov/book/countermeasures/countermeasures-that-work
Strategic Highway Safety Plans:

https://rspcb.safety.fhwa.dot.gov/shspsearch/statesearch.aspx
Intersection Safety Plans: https://safety.fhwa.dot.gov/intersection/stop/fhwasa10010.pdf
Roadway Departure Safety Plans:
https://safety.fhwa.dot.gov/roadway_dept/strat_approach/fhwasa1120/
Human Factors Guidelines: http://www.trb.org/Main/Blurbs/167909.aspx
FHWA Desktop Reference for CRFs:
https://safety.fhwa.dot.gov/speedmgt/ref_mats/fhwasa1304/resources/CRF%20Desktop%20Reference.pdf
NCHRP Report 500 series: http://www.trb.org/Main/Blurbs/152868.aspx

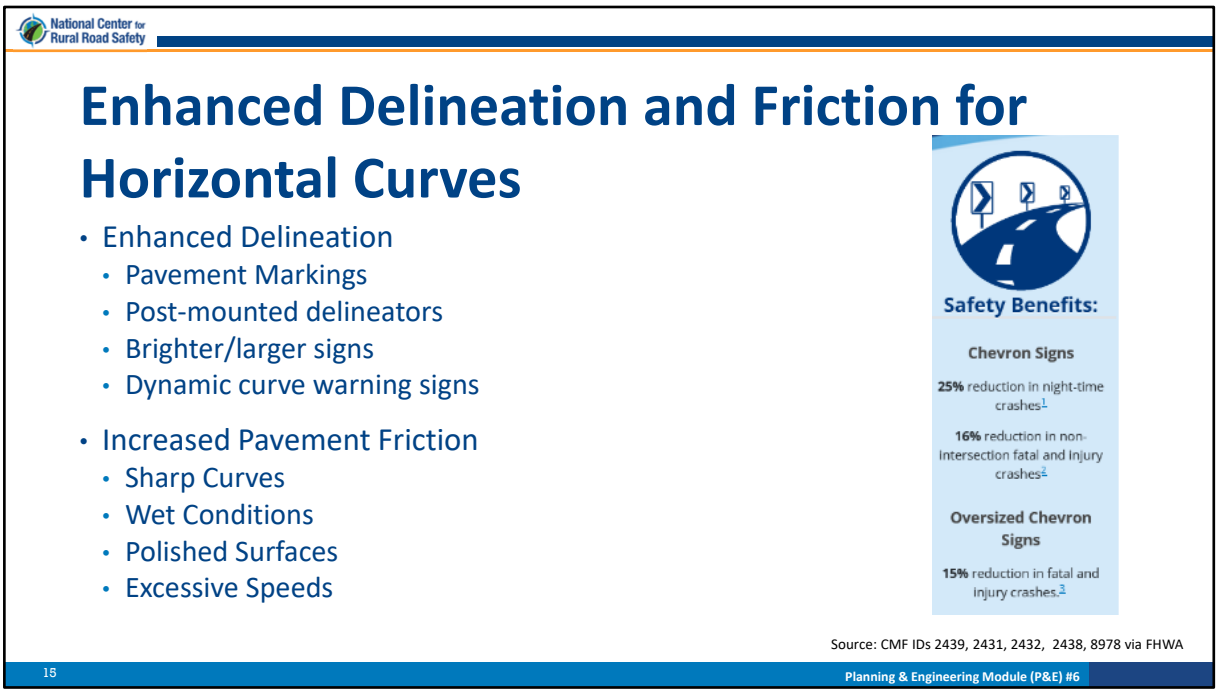


Suggested Comments/Key Message: This slide shows two of the resources where you can find the list of FHWA proven countermeasures, along with the website to access information about the 26 countermeasures divided into themes such as speed management, pedestrian/bicyclist, roadway departures, intersections, and crosscutting.

If your focus crash type was related to pedestrian crashes on urban segments, you may consider looking at the information provided on walkways and medians and pedestrian crossing island in urban and suburban areas.

Reference Materials:
FHWA Proven Safety Countermeasures Website: https://highways.dot.gov/safety/proven-safety-countermeasures
Making Our Roads Safer One Countermeasure at a Time: https://highways.dot.gov/sites/fhwa.dot.gov/files/Proven%20Safety%20Countermeasures%20Booklet.pdf

Proven Safety Countermeasures in Rural Communities: https://highways.dot.gov/sites/fhwa.dot.gov/files/2024-01/FHWA_PSCs_in_Rural_Communities_508.pdf



Suggested Comments/Key Message: Some of the proven safety countermeasures are targeted for systemic application. One of these is enhanced delineation and friction for horizontal curves. These could include countermeasures to enhance delineation such as adding pavement markings, post-mounted delineators, brighter/larger signs or dynamic curve warning signs or to increase pavement friction on sharp curves, curves with polished surfaces or histories of crashes in wet conditions or due to excessive speeds.

As one can see, FHWA provides information on the effectiveness for each of these countermeasures. Additionally, there is a proven safety countermeasure for Systemic Application of Multiple Low-Cost Countermeasures at Stop Controlled Intersections

Reference Materials: n/a
FHWA Proven Safety Countermeasure: Enhanced Delineation and Friction for Horizontal Curves - https://highways.dot.gov/safety/proven-safety-countermeasures/enhanced-delineation-horizontal-curves

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Example: NCHRP 500 Series Countermeasure List

Objectives	Countermeasures	Relative Cost to Implement and Operate	Effectiveness	Typical Timeframe for Implementation
15.1 A: Keep vehicles from encroaching on the roadside	15.1 A1: Install shoulder rumble strips	Low	Tried	Short
	15.1 A2: Install edgelines "profile marking" edgeline rumble strips or modified shoulder rumble strips on section with narrow or no paved shoulders	Low	Experimental	Short
	15.1 A3: Install midlane rumble strips	Low	Experimental	Short
	15.1 A4: Provide enhanced shoulder or delineation and marking for sharp curves	Low	Tried/Prove/ Experimental	Short

Source: NCHRP 500

Suggested Comments/Key Message: Here is an example of some of the data available from NCHRP 500 Series Volume 7: A Guide for Reducing Collisions on Horizontal Curves.

Facilitation Guidance/Interactivity: Present key message and then use the table to describe how this table works. Note the objective, the potential countermeasures, the relative cost, effectiveness and timeframe.

You may want to note that these NCHRP 500 guides are a bit dated, however while the effectiveness and cost may have changed, the recommended countermeasures for the listed objective will likely not have changed.

Reference Materials:
NCHRP 500 Series Volume 7: A Guide for Reducing Collisions on Horizontal Curves
https://www.nap.edu/read/13545/chapter/1

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Exercise: Countermeasure Identification

List any countermeasures you would include in your comprehensive list to address the following crash type and facility:

Focus crash type: Pedestrian crashes

Focus crash facility: Urban signalized intersections

Source: FHWA

Suggested Comments/Key Message: Let's now try our hand at identifying countermeasures. We will have you list any countermeasures you would include in your comprehensive list of countermeasures to address pedestrian crashes at urban signalized intersections. Take a moment and share your thoughts in the chat pod…. *(Review answers placed in chat pod)*

Potential pedestrian countermeasures include: leading pedestrian interval, countdown timers, crosswalks, pedestrian refuges, and increased cycle length for pedestrian crossings.

Facilitation Guidance/Interactivity: Pose the question on the slide and then have them list either through a chat pod (if training is presented virtually) or have them make a list in their participant workbook and then have them share with the group of countermeasures. Then debrief and mention any that were not included from the list.

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Step 3 – Task 2: Evaluate/Screen Countermeasures

- Documented effectiveness
- Implementation and maintenance costs
- Consistency with agency polices, practices and experiences

Suggested Comments/Key Message: Once you have your assembled list of potential countermeasures, ~~in the second task in the countermeasure selection process~~ you will evaluate and screen the initial list, as you will likely have identified more countermeasures that you can discuss at a stakeholder meeting or workshop when determining countermeasures for projects. There are also going to be some countermeasures that are just not consistent with an agency's practices.

You will evaluate the countermeasures based on documented safety effectiveness at reducing the focus crashes, implementation and maintenance costs, and consistency with the agency's policies, practices, and experiences. Implementing countermeasures that are proven to be effective at reducing crashes provides agencies the highest possible level of confidence that their investment will have a positive outcome—a reduction in the focus crashes.

One will often utilize the HSM or CMF Clearinghouse during this step to determine the documented effectiveness. The documented effectiveness along with the implementation and maintenance costs will often be utilized together to determine the benefit cost ratio of the potential countermeasures.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

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Step 3 – Task 3: Select Countermeasures

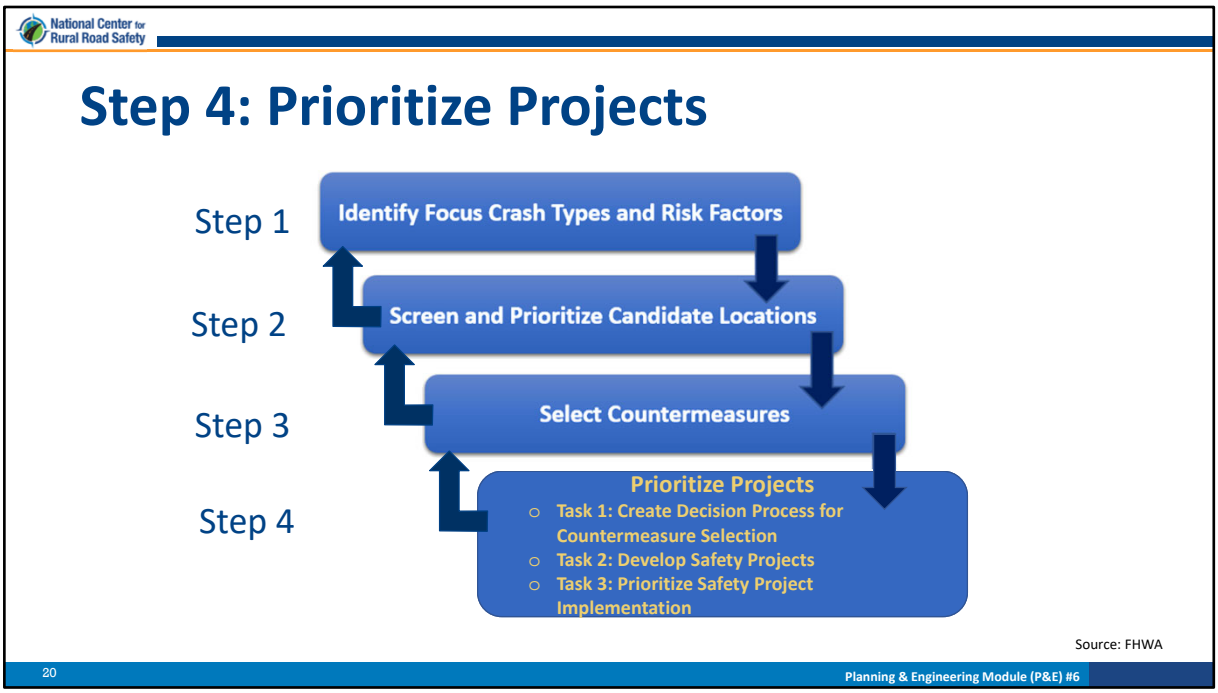
- Represent highest priorities
  - Most cost-effective countermeasures addressing targeted crash types
- Consistent with agency practices and policies
- Provide a range of options for flexibility

Suggested Comments/Key Message: The last task in step three of the systemic safety planning process is to select your countermeasures. This will result in the selection of one or more countermeasures for each focus crash type. These should represent our highest priorities by being the most cost-effective countermeasures that address our targeted crash types while still meeting agencies policies, practices and experiences.

This short list would include primarily low-cost countermeasures, along with a few higher-cost countermeasures. Agencies use this list of counter measures to develop safety projects at specific locations across their roadway system. Although an agency might prefer to deploy one particular countermeasure, there might also be locations where that countermeasure is not an ideal choice (e.g., rumble strips in residential areas). As a result, having alternatives to choose from provides flexibility and acknowledges that all candidate locations in a system are not the same.

I sometimes use the rule – short / intermediate / long term.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf



Suggested Comments/Key Message: The final step in Element 1: the Systemic Safety Planning Process is to prioritize projects. In step 2 we developed a list of prioritized candidate locations and in step 3 a list of countermeasures. In step 4 we will utilize both of those lists to develop a list of high-priority safety improvement projects for implementation. This is again done through three tasks which we will discuss in detail on the following slides.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

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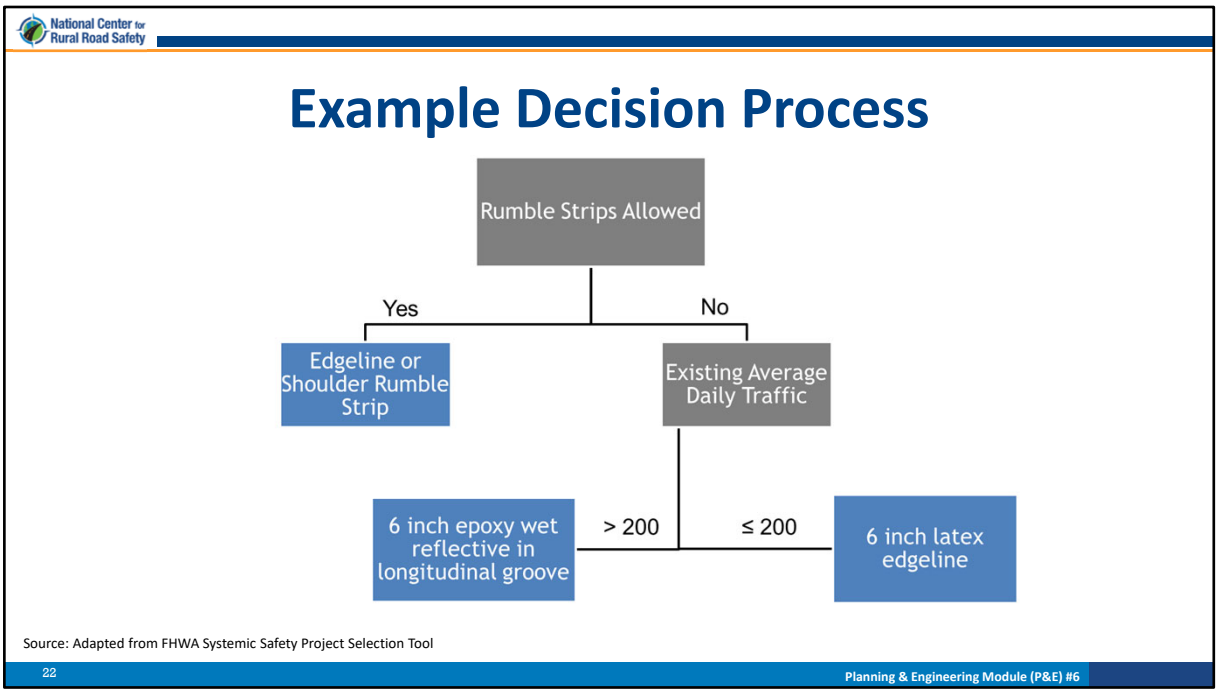
Step 4 – Task 1: Create Decision Process

- **Decision process**: set of criteria to identify appropriate countermeasure.
  - Provides consistency in project development
  - Considers multiple locations for which countermeasures are appropriate and affordable
  - Examples: traffic volume, environment, adjacent land use, or cross-section

Photo Source: FHWA

Suggested Comments/Key Message: In task 1 of step 4, one creates a decision process for each focus crash type to provide a means to consistently assign countermeasures to focus facility locations. A decision process is simply a set of criteria that considers issues such as volume, environment, adjacent land use, or cross-section, and uses them to identify the appropriate countermeasure for high-priority locations. An important distinction of the systemic safety planning process is that the decision-making process does not just identify the most appropriate countermeasure for each location (like when addressing hot spots), it also considers multiple locations with similar crash and risk characteristics to select preferred one or more countermeasures suitable and affordable for widespread implementation.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf



Suggested Comments/Key Message: Here is an example of a decision tree process one could follow when determining how to reduce roadway departure crashes at curves.

In Scenario 1: We have a site that is not near homes and allows for rumble strips. So we follow the tree to the left under "yes" and see that the countermeasure for scenario 1 is to install edgeline or shoulder rumble strips.

For Scenario 2, we have a site that is near homes and therefore rumble strips are not allowed, and the road has an ADT of 300 vpd, what countermeasure is recommended? Take a moment and share your thoughts in the chat pod…. *(Review answers placed in chat pod)*

Correct, in scenario 2, one would use a 6 inch epoxy wet reflective edgeline in a longitudinal groove as the countermeasure.

For Scenario 3, If the same site from scenario 2 had and ADT of 150 vpd on the roadway, what countermeasure is recommended? Take a moment and share your thoughts in the chat pod…. *(Review answers placed in chat pod)*

Correct, in Scenario 3 the recommended countermeasure would be a 6 inch latex edgeline.

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Step 4 – Task 2: Develop Safety Projects

- Apply decision process
- Identify specific countermeasures for each candidate site
- Document decision process and results

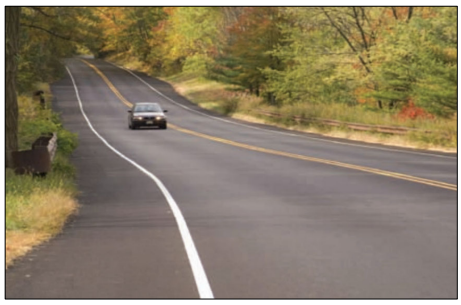

Photo Source: FHWA

Suggested Comments/Key Message: In task 2 you will take the decision process developed in the previous task (i.e., previous slide) and apply it to all your candidate location to determine the recommended countermeasure or countermeasures for each site. As part of this process, you want to make sure you document your decision process and results.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

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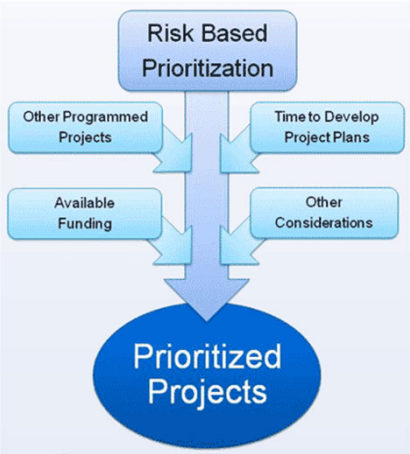
Step 4 – Task 3: Prioritize Safety Projects



Source: FHWA

Suggested Comments/Key Message: Lastly, projects are prioritized for implementation by considering factors such as benefit-cost, funding, other programmed projects, time to develop project plans, expected crash reduction, amount of public outreach needed, and environmental and right-of-way constraints. The prioritized list of locations infers a particular order based on the number of risk factors present at a given location. However, other factors often influence the ability to let a specific project. For example, a project might need to be included in an established capital improvement program or coordinated with other projects/programs.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

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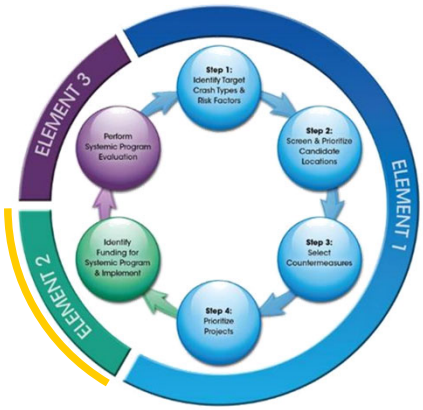
Element 2: Funding – Program and Implementation

3 Elements:

1. Systemic Safety Planning Process
2. **A Framework for Balancing Systemic and Traditional Safety Investments**
3. Evaluation of a Systemic Safety Program



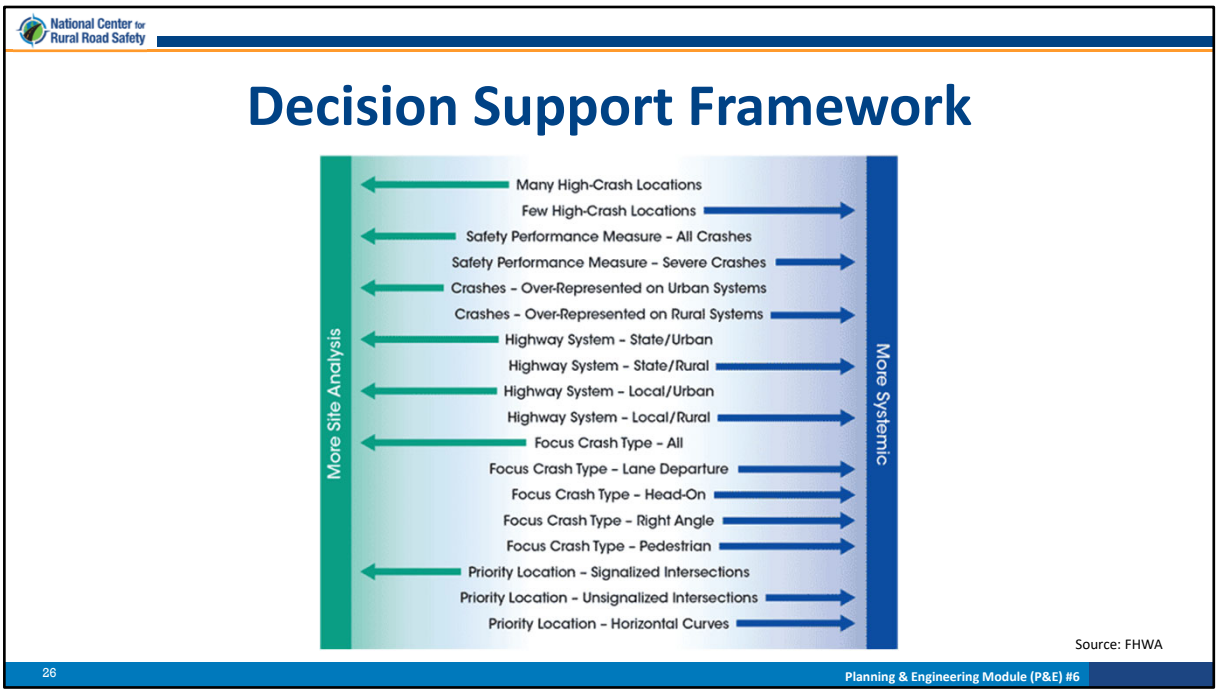
Source: FHWA

Suggested Comments/Key Message: Agencies interested in adding a systemic component to their highway safety program quickly realize their next decision involves determining how to distribute their safety investments among projects identified through the traditional site analysis approach and projects identified through the systemic approach. Element 2 of the Systemic Safety Project Selection Tool provides a framework determining an appropriate balance of safety investments between the site analysis and systemic approaches

This framework is made up of the following three components:

- Review past funding practices to provide insight into historical safety spending patterns.
- Apply the funding determination framework to determine the appropriate balance of safety investments.
- Assess the potential benefit to be gained by investing funding in systemic

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf



Suggested Comments/Key Message: A funding determination framework assumes two key points:

- First, there is no suggestion or expectation that any roadway agency's safety program will be 100 percent oriented toward deployment of systemic projects.
- This framework suggests, however, that in every agency there will be some balance between projects derived from the site analysis approach and projects derived from the systemic approach.

Agencies should consider a variety of crash and roadway characteristics to determine the balance of safety investments between site analysis and systemic projects. Such considerations suggest a general balance along a safety funding continuum, with one end representing all site analysis projects and the other end representing all systemic projects. Agencies can develop a continuum with characteristics appropriate for their system. The graphic on the slide illustrates a continuum using recommended characteristics for the funding determination framework.

Using this graphic, an agency may decide what approach to put an emphasis on. For example, let's look at the third line. If our safety performance measure looks at all crashes, we would want to include more site analysis, while if our safety performance measure looks at severe crashes, we would want to include more systemic analysis as these crash

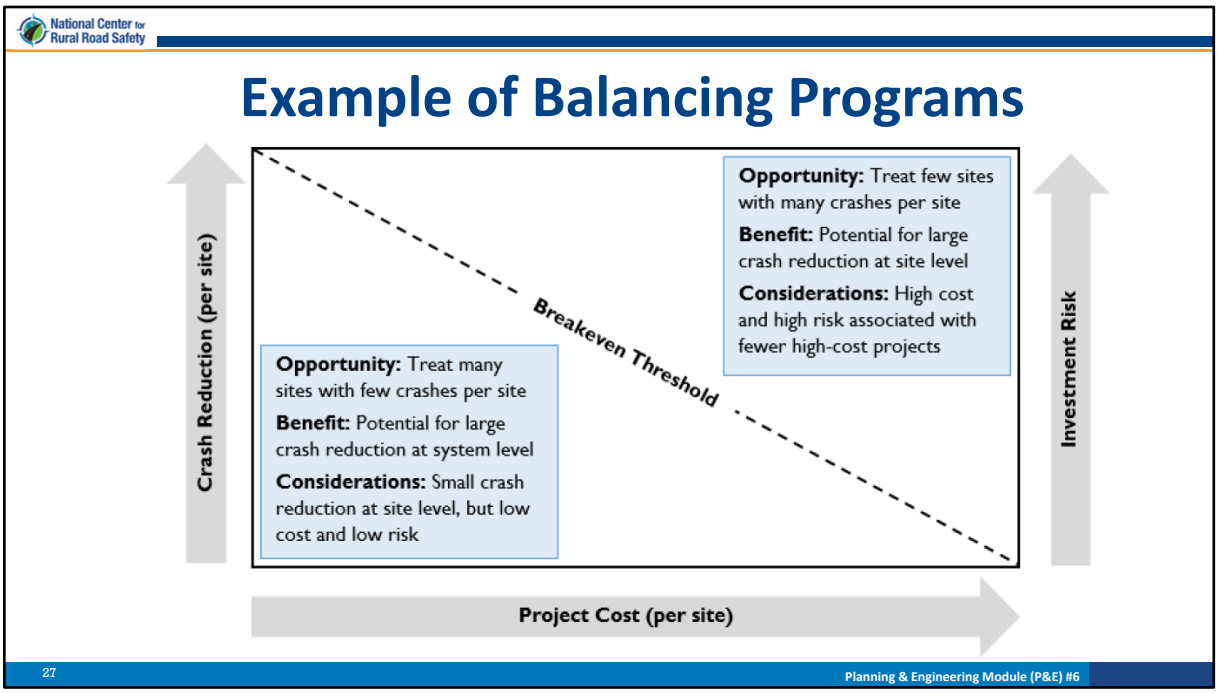
types tend to be more widespread.

Facilitation Guidance/Interactivity: Use the graphic on the slide to discuss how an agency may decide what approach to put an emphasis on. If time permits, you could propose the Q&A in the section below to allow participants the opportunity to apply the framework.

Potential Answers/ Sample Q&A:
Ask: If you are a local agency with mostly rural roads, which type of projects should you place more of an emphasis on?
Answer: More systemic projects.

Ask: However, if you are a local agency with more urban roads instead, what type of projects should you place more of an emphasis on?
Answer: More site analysis projects.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf



Suggested Comments/Key Message: The second element also deals with implementing projects. This graphic is a good tool used to determine when a systemic based approach is a better option and when a site-specific approach is merited. As you can see here, as the crash reduction per site increases, the site-specific approach proves to be more beneficial; however, as the project cost per site increases the investment risk increases.

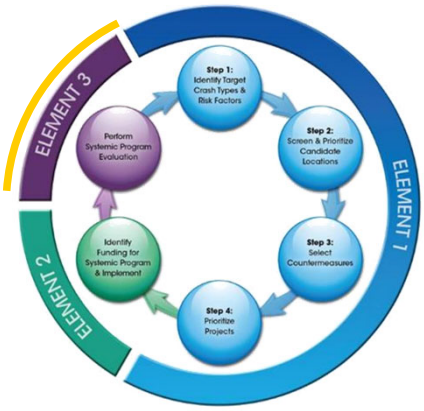
This graphic also summarizes the benefits of the two approaches as well as any considerations for each approach.

National Center for Rural Road Safety

Element 3: Systemic Safety Planning Process

3 Elements:

1. Systemic Safety Planning Process
2. A Framework for Balancing Systemic and Traditional Safety Investments
3. **Evaluation of a Systemic Safety Program**



Source: FHWA

Suggested Comments/Key Message: Evaluation of a planning process such as the systemic approach is critical to continue to spend funds where high crash rates or frequencies may not be linked. As risk factors change and focus crash types become more or less important, shifts should rightly occur in the program. Reduction in crashes along treated segment or intersections should be highlighted and successes should be shared. This is accomplished through Element 3 of the Systemic Safety Planning Process.

This element introduces the safety performance evaluation process for a systemic safety program. As the systemic safety evaluation process is new and continually evolving, the tool document does not present a process or a framework for evaluation. Instead, it provides an overview of an approach and potential methods, including the data needs and performance measures for these methods. The guide also discusses several scenarios agencies might face as they evaluate their program. As the systemic approach to safety evolves and implementation continues, additional research is necessary to confirm systemic safety evaluation techniques.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

National Center for Rural Road Safety

Three-Level Systemic Program Evaluation

1. Output / activity level
   - Are projects implemented?
2. Observed trends in crash frequency and severity for focus crash type
   - Are trends decreasing?
3. Countermeasure performance
   - What is actual effectiveness?

Suggested Evaluation of Data

Countermeasures
- Definition of improvement implemented at site
- Precise implementation location
- Precise implementation data

Crash Data
- Severe crash data for focus crash type before implementation
- Severe crash data for focus crash type after implementation

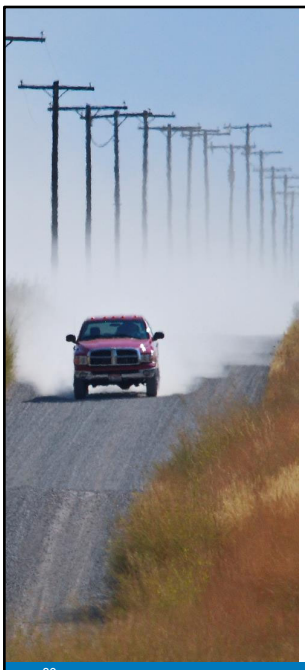
Before and After Site Conditions
- Roadway or intersection geometry
- Intersection traffic control device
- Shoulder surface width and type
- Road division
- Median width and type
- Speed limit
- Average daily traffic volumes
- Average daily entering vehicles
- Roadway classification
- Area type (rural/urban/suburban)

Suggested Comments/Key Message: A systemic safety program's effectiveness can be measured by reduced system risk. Depending on the availability of data applied in the project identification process, risk might not be easily quantifiable. Instead, changes in the number of severe focus crash types, especially on the focus facility types, become the long-term performance measurement.

The recommended systemic safety evaluation process occurs at three levels, as follows:

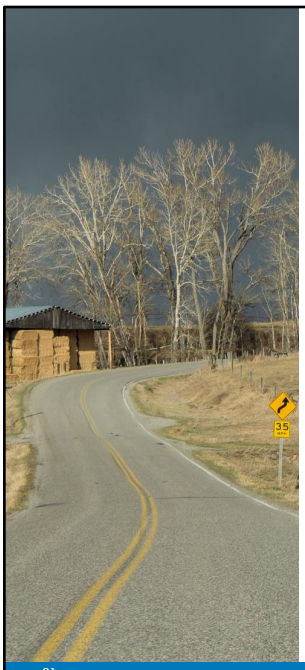
- *Output:* What is the output of the systemic safety program? Is the systemic safety program being implemented as planned and programmed? Are high priority countermeasures being deployed at the right type of locations and at the number of locations planned?
- *Focus Crash Type*: With the systemic application of low-cost safety improvements, one wants to ensure if they effectively reduced the identified focus crash types? That is, are the severe crashes trending down?
- *Countermeasure Performance*: Within each crash type, are deployed countermeasures performing as expected? In this case you would look at the performance of each individual low-cost safety improvement.

Reference Materials: n/a



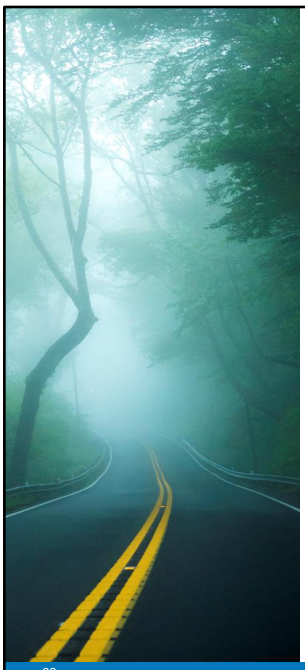
Review: Learning Outcomes

- List the differences between systemic screening and other network screening methods
- Describe the three elements that make up the Systemic Safety Project Selection Tool
- Describe and apply the 4-step systemic planning process
- Identify resources that are helpful in conducting systemic safety analysis



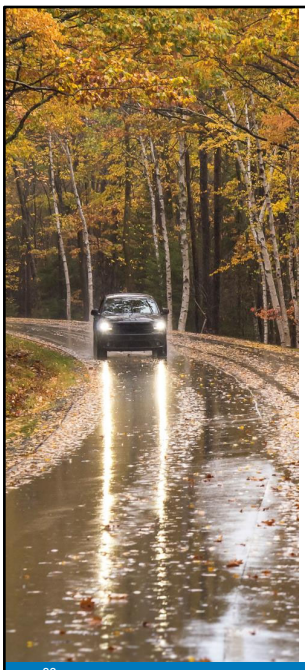
Resources Part 1

- **National Center for Rural Road Safety**
  https://ruralsafetycenter.org/
- **National LTAP & TTAP Association**
  https://nltapa.org/
- **National Association of County Engineers**
  https://www.countyengineers.org/
- **West Region Transportation Workforce Center**
  https://wrtwc.org/
- **FHWA Local Rural Road Safety Program**
  https://highways.dot.gov/safety/local-rural



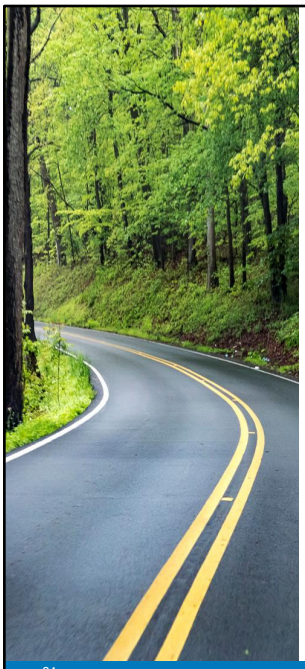
Resources Part 2

- **Strategic Highway Safety Plans**
  https://highways.dot.gov/safety/hsip/shsp/shsp-resources
- **Highway Safety Manual (HSM)**
  http://www.highwaysafetymanual.org/Pages/default.aspx
- **NCHRP Report 500 Series**
  http://www.trb.org/Main/Blurbs/152868.aspx
- **AASHTOWare Safety Analyst**
  https://www.aashtoware.org/products/safety/safety-overview/
- **US Road Assessment Program (usRAP)**
  http://www.usrap.org/



Resources Part 3

- **Systemic Safety Project Selection Tool**
  https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf
- ***NEW* Systemic Safety User Guide**
  https://highways.dot.gov/safety/data-analysis-tools/systemic/systemic-safety-user-guide
- **Crash Tree Maker and User Guide**
  https://highways.dot.gov/safety/local-rural/local-road-safety-plans (systemic tools drop down)
- **CMF Clearinghouse**
  http://www.cmfclearinghouse.org/



Resources Part 4

- **Reliability of Safety Management Methods: Systemic Safety Programs**
  https://safety.fhwa.dot.gov/rsdp/downloads/fhwasa16041.pdf
- **Highway Safety Benefit-Cost Analysis Guide and Tool**
  https://highways.dot.gov/safety/data-analysis-tools/rsdp/rsdp-tools/highway-safety-benefit-cost-analysis-guide-and-tool
- **FHWA Systemic Safety Website**
  https://highways.dot.gov/safety/data-analysis-tools/systemic